INCH-POUND

MIL-R-6106/28F
30 November 1994
SUPERSEDING
MIL-R-6106/28E
12 July 1993

MILITARY SPECIFICATION SHEET

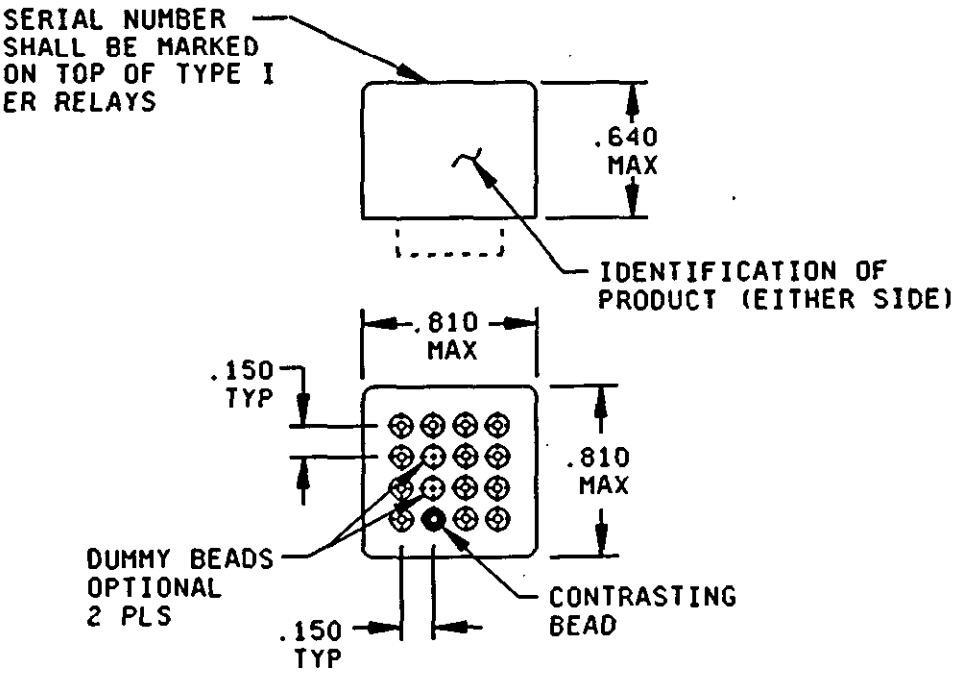
RELAYS, ELECTROMAGNETIC (INCLUDING ESTABLISHED RELIABILITY (ER) TYPES), TYPE I, 4 PDT, LOW LEVEL TO 5 AMPERES, PERMANENT MAGNET DRIVE, HERMETICALLY SEALED

(F)

INACTIVE FOR NEW DESIGN AS OF 30 NOVEMBER 1994. THIS SPECIFICATION AND ITS ASSOCIATED QUALIFIED PARTS LIST ARE SCHEDULED FOR CANCELLATION IN OCTOBER 1995. REFER TO TABLE V AND MIL-R-83536 FOR SUPERSESSION DATA.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-6106.



CONFIGURATION A

FIGURE 1. Dimensions and configurations.

(F) denotes changes

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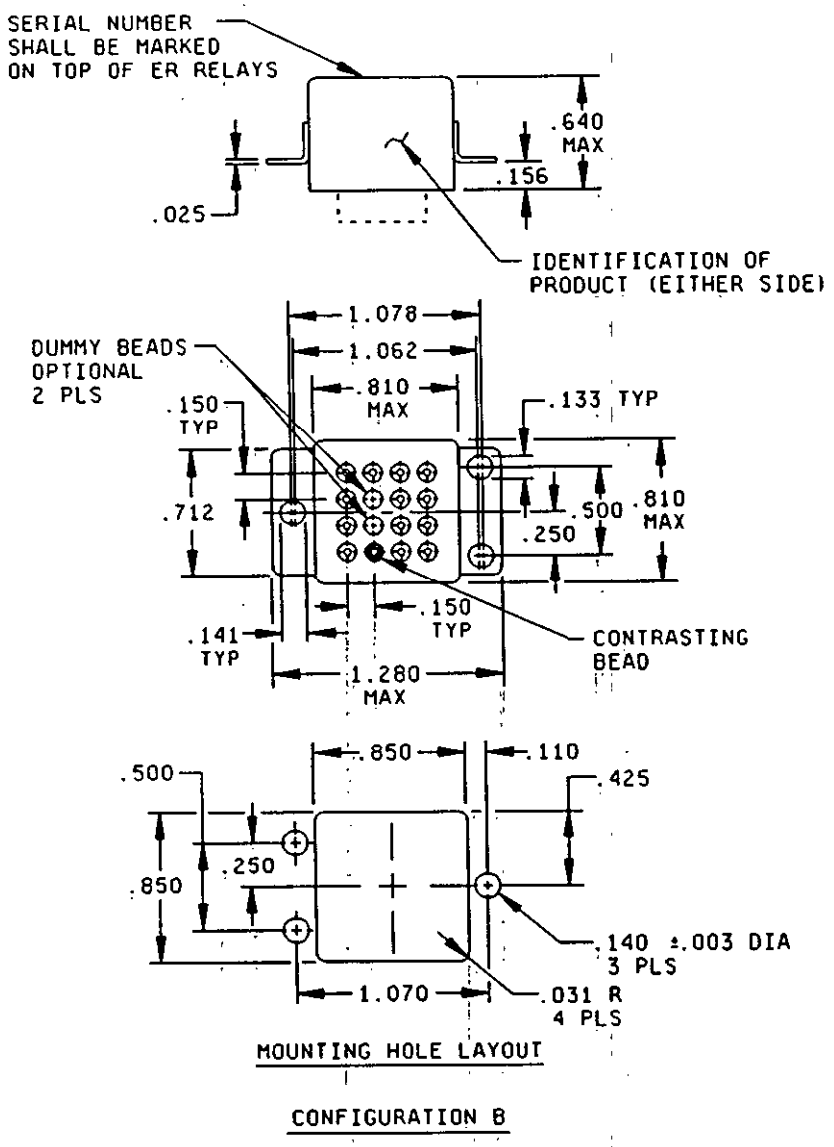
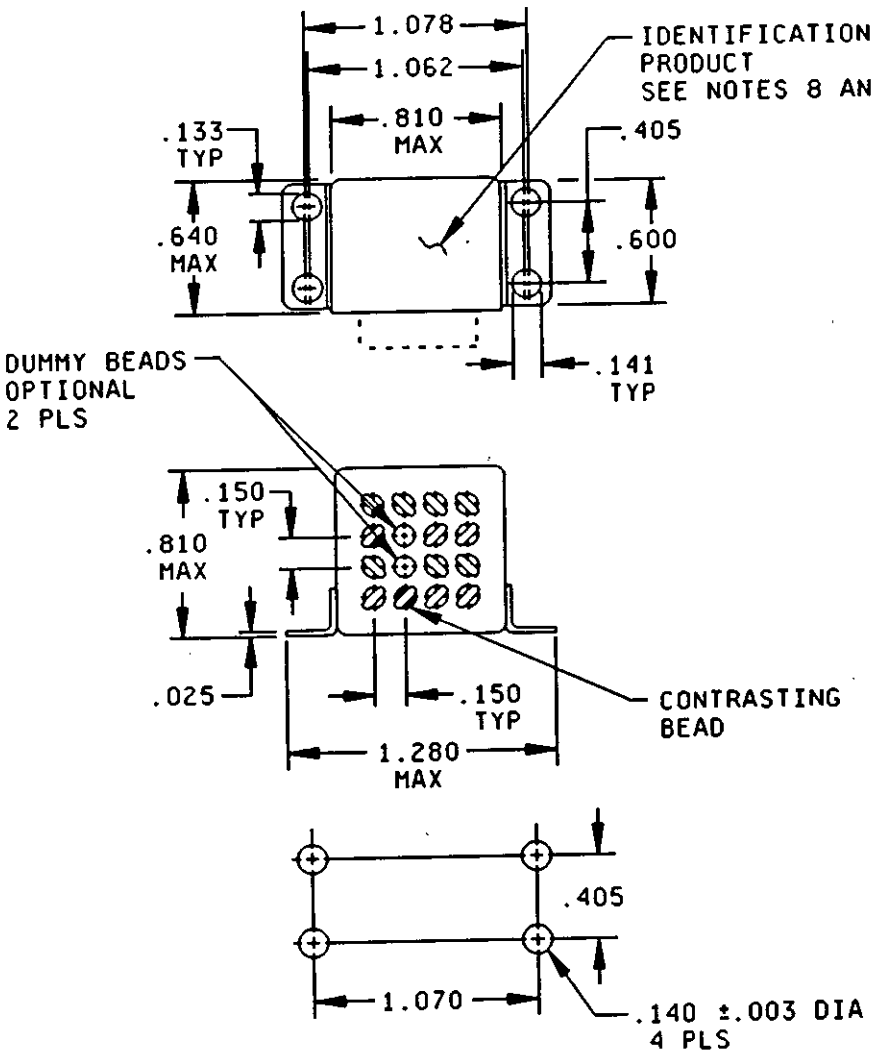


FIGURE 1. Dimensions and configurations - Continued.

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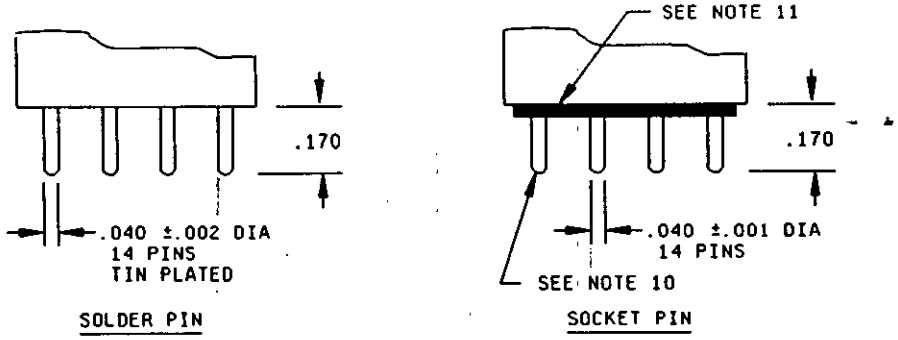


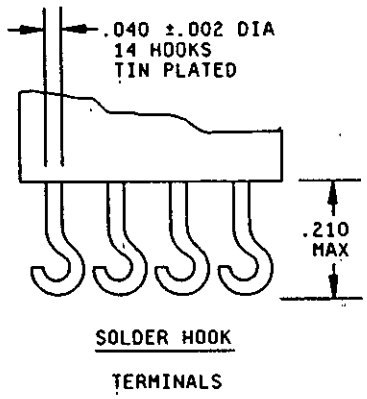
MOUNTING HOLE LAYOUT

CONFIGURATION C

FIGURE 1. Dimensions and configurations - Continued.

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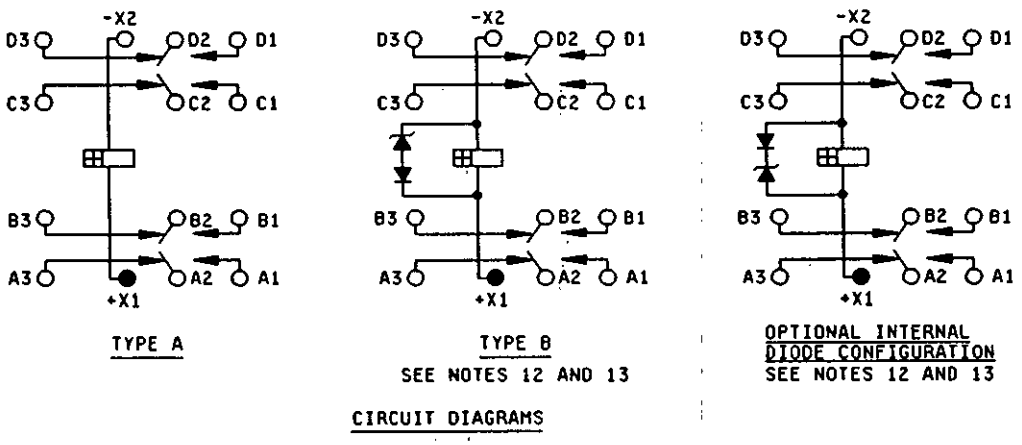


FIGURE 1. Dimensions and configurations - Continued.

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Inches	mm	Inches	mm	Inches	mm
.001	0.03	.140	3.56	.425	10.80
.002	0.05	.141	3.58	.500	12.70
.003	0.08	.150	3.81	.640	16.26
.025	0.64	.156	3.96	.712	18.08
.031	0.76	.170	4.32	.810	20.57
.040	1.02	.210	5.33	.850	21.59
.110	2.79	.250	6.35	1.062	26.97
.133	3.38	.405	10.29	1.078	27.38
				1.280	32.51

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 inch (0.25 mm).
4. There shall be affixed to the relay a legible circuit diagram that identifies each terminal location specified.
5. These relays are polarized monostable.
6. This relay shall not operate or be damaged by reverse polarity. Semiconductors shall not be used for this purpose.
7. Permanent magnet drive consists of a permanent magnet with its flux path switched and combined with the electromagnetic flux.
8. Applicable to configuration C only. The circuit diagram, manufacturer's PIN, and the military PIN shall be marked on the near side. The remaining portion of the nameplate data shall be marked on the far side.
9. Applicable to configuration C only. Relays shall be marked with the manufacturer's name or source code and data code. In addition, ER relays shall be marked with the serial number. Marking shall be with the bottom of print adjacent to the near side.
10. Socket pin terminals shall be gold-plated in accordance with MIL-G-45204, type II, class I. The nickel underplating shall be 50 to 150 microinches thick (minimum).
11. Silicone rubber gasket, AMS 3332, shore hardness 25 ±5, thickness .050 ±.005 inch (1.27 ±0.13 mm).
12. Back EMF (transient voltage): 42 V dc (maximum).
13. JANTX or equivalent screened semiconductors shall be used in ER relays with internal coil suppression. Relays using suppression devices shall continue to operate should the suppression circuit be in a failure mode. Diodes shall have a peak inverse voltage of 600 V dc (minimum).

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

CONTACT DATA:

Life: Unless otherwise specified, 100,000 cycles.

Load ratings:

High level (relay case grounded):

Resistive: 5 amperes at 28 V dc; 115 V ac, 400 Hz, 1 phase; 115/200 V ac, 400 Hz, 3 phase.

Inductive: 3 amperes at 28 V dc; 5 amperes at 115 V ac, 400 Hz, 1 phase and 115/200 V ac, 400 Hz, 3 phase. (Life: 20,000 cycles.)

Motor: 2 amperes at 28 V dc; 3 amperes at 115 V ac, 400 Hz, 1 phase and 115/200 V ac, 400 Hz, 3 phase.

Lamp: 1 ampere at 28 V dc and 115 V ac, 400 Hz, 1 phase.

Low level: 10 to 50 μA at 10 to 50 mV dc or peak ac. 1/ 2/ 3/

Intermediate current: Applicable.

Transfer load: Not applicable.

Mechanical life (reduced current): 1.25 amperes at 28 V dc; 115 V ac, 1 phase, 400 Hz; and 115/200 V ac, 3 phase, 400 Hz. (Life: 400,000 cycles.)

Mixed loads: Applicable. 4/

Contact voltage drop or resistance: 1/ 2/ 3/

High level:

Initial: 0.150 volt maximum.
After life: 0.175 volt maximum.

Low level:

Initial: 0.050 ohm maximum.
After life: 0.150 ohm maximum.

Contact bounce: 0.5 millisecond maximum.

Break bounce (normally open contacts only): 0.1 millisecond maximum (applicable to DC suppressed coils only).

1/ For low level, the following shall apply:
   a. During endurance and operational reliability testing, contact load shall be 10 to 50 microamperes; 10 to 50 millivolts open circuit voltage, 100 ohms maximum contact resistance.
   b. Static contact resistance shall be performed at 50 mA maximum, 50 mV maximum.

2/ For all ER relays, the alternate low level test of operational reliability shall be used.

3/ For group A contact voltage drop test, high level testing shall be performed first, followed by low level testing. The contacts shall not make or break the high level load.

4/ Relay shall be capable of switching low level while switching any of the other rated loads on adjacent contacts.

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Overload current: 20 amperes dc.
30 amperes ac.

Rupture current: 25 amperes dc.
40 amperes ac.

Time current relay characteristics: Applicable (see table I).

TABLE I. Time current relay characteristics. 1/ 2/

1	10 A	1 hour
2	15 A	20 seconds
3	37.5 A	2 seconds
4	75 A	.53 seconds

1/ CAUTION: Compare with time current characteristics of the associated circuit protective devices.

2/ All relays shall withstand overload and fault currents. Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of four different current levels in the sequence listed in table I. Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz, 3 phase. Cooling time between successive applications shall be 30 minutes. Tests shall be performed on both normally open and normally closed contacts. There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion.

COIL DATA 5/: See table II.

Operate time: 6 milliseconds maximum with rated coil voltage.

Release time: 6 milliseconds maximum from rated coil voltage.

Duty rating: Continuous.

5/ CAUTION: Due to possible interaction of relay magnetic fields, the following spacing requirements, as a minimum, shall be considered in dense packaging situations:
a. Row to row assisting fields .125 inch.
b. Row to row opposing fields .1875 inch.
c. Side to side alternating fields .0625 inch.
d. Side to side like fields .125 inch.

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TABLE II. Dash numbers and characteristics. 1/

PIN M6106/28- 2/ 3/	Type	Terminal type	Mounting configuration	Circuit type	Coil data								
					Rated voltage (V dc) 4/	Resistance Ω ±10% at 25°C	Maximum		Pickup volt (max) (V dc)			Hold voltage (V dc) 6/	Drop-out voltage (V dc) 6/
							Voltage (V dc) 5/	Current (A)	Normal 6/	High temp test	Continuing current test		
001 7/	I	Solder pin	A	A	28	400	29	.081	18	19.8	22.5	7.0	1.5
002	I ER	"	"	A	"	"	"	"	"	"	"	"	"
003 7/	I	"	"	B	"	"	"	"	"	"	"	"	"
004	I ER	"	"	B	"	"	"	"	"	"	"	"	"
005 7/	I	Solder hook	"	A	"	"	"	"	"	"	"	"	"
006	I ER	"	"	A	"	"	"	"	"	"	"	"	"
007 7/	I	"	"	B	"	"	"	"	"	"	"	"	"
008	I ER	"	"	B	"	"	"	"	"	"	"	"	"
009 7/	I	Socket pin	"	A	"	"	"	"	"	"	"	"	"
010	I ER	"	"	A	"	"	"	"	"	"	"	"	"
011 7/	I	"	"	B	"	"	"	"	"	"	"	"	"
012	I ER	"	"	B	"	"	"	"	"	"	"	"	"
013 7/	I	Solder pin	B	A	"	"	"	"	"	"	"	"	"
014	I ER	"	"	A	"	"	"	"	"	"	"	"	"
015 7/	I	"	"	B	"	"	"	"	"	"	"	"	"
016	I ER	"	"	B	"	"	"	"	"	"	"	"	"
017 7/	I	Solder hook	"	A	"	"	"	"	"	"	"	"	"
018	I ER	"	"	A	"	"	"	"	"	"	"	"	"
019 7/	I	"	"	B	"	"	"	"	"	"	"	"	"
020	I ER	"	"	B	"	"	"	"	"	"	"	"	"
021 7/	I	Socket pin	"	A	"	"	"	"	"	"	"	"	"
022	I ER	"	"	A	"	"	"	"	"	"	"	"	"
023 7/	I	"	"	B	"	"	"	"	"	"	"	"	"
024	I ER	"	"	B	"	"	"	"	"	"	"	"	"
025 7/	I	Solder pin	C	A	"	"	"	"	"	"	"	"	"
026	I ER	"	"	A	"	"	"	"	"	"	"	"	"
027 7/	I	"	"	B	"	"	"	"	"	"	"	"	"
028	I ER	"	"	B	"	"	"	"	"	"	"	"	"
029 7/	I	Solder hook	"	A	"	"	"	"	"	"	"	"	"
030	I ER	"	"	A	"	"	"	"	"	"	"	"	"
031 7/	I	"	"	B	"	"	"	"	"	"	"	"	"
032	I ER	"	"	B	"	"	"	"	"	"	"	"	"
037	I	Solder pin	B	A	12	100	14.5	.161	9	9.9	11.25	4.5	.75
038	I	Solder pin	"	B	"	"	"	"	"	"	"	"	"
039	I	Solder hook	"	A	"	"	"	"	"	"	"	"	"
040	I	Solder hook	"	B	"	"	"	"	"	"	"	"	"
041	I	Socket pin	"	A	12	100	14.5	.161	9	9.9	11.25	4.5	.75
042	I	Socket pin	"	B	"	"	"	"	"	"	"	"	"
043	I	Solder pin	C	A	"	"	"	"	"	"	"	"	"
044	I	Solder pin	"	B	"	"	"	"	"	"	"	"	"
045	I	Solder hook	"	A	"	"	"	"	"	"	"	"	"
046	I	Solder hook	"	B	"	"	"	"	"	"	"	"	"

1/ Each relay has high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications.

2/ For type I ER relays, a suffix letter M, U, X, or P, to indicate the applicable failure rate level, shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): M = 1 percent, U = 0.5 percent, X = 0.3 percent, and P = 0.1 percent. Examples: M6106/28-002M and M6106/28-032P.

3/ M6106/28-033 through M6106/28-036, -047, and -048 are canceled without replacement.

4/ CAUTION: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.

5/ When maximum ambient temperature does not exceed +85°C, the maximum coil voltage shall be 32 V dc.

6/ Over the temperature range.

7/ Type I relays are inactive for new design after 14 December 1990. See table IV for cross-reference data.

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ELECTRICAL DATA:

Insulation resistance, initial: 100 megohms.

After life or environmental tests: 50 megohms.

Dielectric withstanding voltage (sea level):	Initial V rms (60 Hz)	After life tests V rms (60 Hz)
Coil to case	1,000	750
All other points	1,000	750

Dielectric withstanding voltage (altitude) 6/:	80,000 ft. V rms (60 Hz)	300,000 ft. V rms (60 Hz)
Coil to case	250	500
All other points	250	500

ENVIRONMENTAL CHARACTERISTICS:

Temperature range: -70°C to +125°C.

Maximum altitude rating: 300,000 feet.

Acceleration: Applicable (15 g's maximum).

Shock (specified pulse): MIL-STD-202, method 213, test condition C (100 g's), except configurations A and B peak g value shall be 200 g's. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum for open contacts.

Vibration (sinusoidal): Configurations A and B shall be 30 g's, 70 to 3,000 Hz. Configuration C shall be 20 g's, 57 to 3,000 Hz. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum for open contacts.

Vibration (random): MIL-STD-202, method 214, for configurations A and B, test condition IG (0.4 g^2/Hz); for configuration C, test condition IE (0.2 g^2/Hz). Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum for open contacts.

PHYSICAL DATA:

Terminal strength:

Solder hook terminals:

Twist test: Not applicable.

Pull force: 5 ±.5 pounds.

Solder pin terminals:

Bend test: Applicable.

Pull force (0.040 diameter): 5 ±0.5 pounds.

Plug-in terminals:

Bend test: Applicable

Pull force (0.040 diameter): 5 ±0.5 pounds.

6/ Dielectric may be improved by suitable insulation of terminals and wiring after installation.

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Qualification by similarity: See table III.

TABLE III. Qualification by similarity. 1/

PIN M6106/28-	Loads		Dynamics			Environmental
	Basic qualification	ER				
	A	D	A	B	C	A
	DC 2/ 3/	DC 2/ 3/	Configuration A 2/ 3/	Configuration B 2/ 3/	Configuration C 2/ 3/	DC 2/ 4/
001 5/	4	3	2			4
002 "	4	3	2			4
003 "	1, 2, 3	1, 2	1			1, 2, 3
004 "	2, 3	1, 2	1			2, 3
005 "	4	3	2			4
006 "	4	3	2			4
007 "	1, 2, 3	1, 2	1			1, 2, 3
008 "	2, 3	1, 2	1			2, 3
009 "	4	3	2			4
010 "	4	3	2			4
011 "	1, 2, 3	1, 2	1			1, 2, 3
012 "	2, 3	1, 2	1			2, 3
013 "	4	3		2		4
014 "	4	3		2		4
015 "	1, 2, 3	1, 2		1		1, 2, 3
016 "	2, 3	1, 2		1		2, 3
017 "	4	3		2		4
018 "	4	3		2		4
019 "	1, 2, 3	1, 2		1		1, 2, 3
020 "	2, 3	1, 2		1		2, 3
021 "	4	3		2		4
022 "	4	3		2		4
023 "	1, 2, 3	1, 2		1		1, 2, 3
024 "	2, 3	1, 2		1		2, 3
025 "	4	3			2	4
026 "	4	3			2	4
027 "	1, 2, 3	1, 2			1	1, 2, 3
028 "	2, 3	1, 2			1	2, 3
029 "	4	3			2	4
030 "	4	3			2	4
031 "	1, 2, 3	1, 2			1	1, 2, 3
032 "	2, 3	1, 2			1	2, 3
037 6/	4	3		2		4
038 "	1, 2, 3	1, 2		1		1, 2, 3
039 "	4	3		2		4
040 "	1, 2, 3	1, 2		1		1, 2, 3
041 "	4	3		2		4
042 "	1, 2, 3	1, 2		1		1, 2, 3
043 "	4	3			2	4
044 "	1, 2, 3	1, 2			1	1, 2, 3
045 "	4	3			2	4
046 "	1, 2, 3	1, 2			1	1, 2, 3

See footnotes on next page.

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TABLE III. Qualification by similarity - Continued. 1/

1/ See amendment 1, MIL-R-6106, for guidelines on determining how relays are to be grouped and ranked within each subgroup.
2/ DC suppressed coils: Discretes qualify only discretes; hybrids qualify only hybrids.
3/ Terminals: Socket pin type terminals are considered worst cases. Solder pins and hooks may be qualified by similarity.
4/ Socket pins (with gasket seals) are considered nonsimilar to solder pins and hooks and must be tested separately.
5/ Similarity exists only between dash numbers 001 and 032 (28 V dc coils).
6/ Similarity exists only between dash numbers 037 through 048 (12 V dc coils).

Terminal solderability: Applicable to solder pin and solder hook terminals only.

Dimensions and configurations: See figure 1.

Weight (maximum): .058 pound (26.4 grams).

Seal: Hermetic; relays shall be sealed by welding (laser, TIG (tungsten inert gas) or other suitable means as approved by the qualifying activity).

Construction (internal and external): All welded, except that coil magnet wire to coil lead wire shall be soldered.

Identification of product: Applicable.

Quality conformance inspection: Performance of group C inspection for Type I and Type I-ER relays is not required. Performance of Retention of Qualification for Type I-ER relays is not required, provided the manufacturer possesses and maintains a current qualification to MIL-R-83536.

(F) Cross-reference for Government logistics support: See table IV.

(F) TABLE IV. Cross-reference for Government logistical support.

PIN M6106/28-	Support with PIN M83536	PIN M6106/28-	Support with PIN M83536
001	/5-017	024	/6-022
002	/5-017	025	/5-023
003	/6-017	026	/5-023
004	/6-017	027	/6-023
005	/5-018	028	/6-023
006	/5-018	029	/5-024
007	/6-018	030	/5-024
008	/6-018	031	/6-024
009	/5-019	032	/6-024
010	/5-019	033	1/
011	/6-019	034	1/
012	/6-019	035	1/
013	/5-020	036	1/
014	/5-020	037	/5-012
015	/6-020	038	/6-012
016	/6-020	039	/5-013
017	/5-021	040	/6-013
018	/5-021	041	/5-014
019	/6-021	042	/6-014
020	/6-021	043	/5-015
021	/5-022	044	/6-015
022	/5-022	045	/5-016
023	/6-022	046	/6-016
		047	1/
		048	1/

1/ M6106/28-033 through -036; -047; and -048 are canceled without replacement.

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(F) Supersession data: See table V.

(F) TABLE V. Supersession data.

Superseded PIN M6106/28-	Replacement PIN M83536	Superseded PIN M6106/28-	Replacement PIN M83536
001	/5-017	024	/6-022
002	/5-017	025	/5-023
003	/6-017	026	/5-023
004	/6-017	027	/6-023
005	/5-018	028	/6-023
006	/5-018	029	/5-024
007	/6-018	030	/5-024
008	/6-018	031	/6-024
009	/5-019	032	/6-024
010	/5-019	033	1/
011	/6-019	034	1/
012	/6-019	035	1/
013	/5-020	036	1/
014	/5-020	037	/5-012
015	/6-020	038	/6-012
016	/6-020	039	/5-013
017	/5-021	040	/6-013
018	/5-021	041	/5-014
019	/6-021	042	/6-014
020	/6-021	043	/5-007
021	/5-022	044	/6-015
022	/5-022	045	/5-008
023	/6-022	046	/6-016
		047	1/
		048	1/

1/ M6106/28-033 through -036; -047; and -048 are canceled without replacement.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-0940-02)

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